

Every mark with body-independent noise is a momentum-transfer mark

Theorem A of the mark-cost note was proved for the von Neumann coupling $\lambda\hat{y}\hat{P}_A$. Among marks whose noise does not depend on the body, that class is all there is. For an arbitrary unitary coupling of the body to an apparatus and an arbitrary pointer, suppose the reading error, the delivered impulse and the position disturbance are operators of the apparatus alone, independent of the body's state. Then:

1. the error $\hat{N}$ and the impulse $\hat{D}$ are canonically conjugate, $[\hat{N}, \hat{D}] = -i\hbar$, and the position disturbance $\hat{X}$ commutes with both (Proposition 1);
2. if the mark leaves the position undisturbed, $\hat{X} = 0$, the coupling is $e^{-i\hat{y}\otimes\hat{B}/\hbar}$ followed by an apparatus unitary, with the pointer canonically conjugate to $\hat{B}$: Theorem A's form exactly, with $\hat{B}$ in place of $\lambda\hat{P}_A$ and extra apparatus degrees of freedom allowed (Proposition 2);
3. the recoil bound $s\sum_j\Delta_j \geq 8(1-2\epsilon)\hbar$ of the recoil note holds for every protocol of such marks, including those that disturb position (Proposition 3).

This is option 1 of the plan the user set on 2026-09-22 for extending Theorem A. The class is Ozawa's measurements with *independent intervention*, for which he proves the Heisenberg product $\epsilon\eta \geq \hbar/2$ (PRA **67**, 042105, 2003; Ann. Phys. **311**, 350, 2004; metadata); the joint-measurement analogue is Arthurs and Goodman (PRL **60**, 2447, 1988, metadata). Proposition 2 is a unitary-dilation form of the classification of translation-covariant instruments on the line (Davies, *Quantum Theory of Open Systems*, 1976, metadata), stated here in the language of marks. What is new is the transfer to the Galileo comparison, Proposition 3. The class needs a pointer with continuous, unbounded spectrum, since the pointer and the impulse form a canonical pair; marks with a discrete or bounded pointer lie outside it and are covered by the disturbance note. Exploratory; no ledger promotion.

1. The identity

A mark is a unitary U on body $\otimes$ apparatus, applied impulsively, followed by reading a pointer $\hat{R}$ of the apparatus. In the Heisenberg picture define

$$\hat{N} = U^\dagger \hat{R} U - \hat{y}, \quad \hat{D} = U^\dagger \hat{p} U - \hat{p}, \quad \hat{X} = U^\dagger \hat{y} U - \hat{y},$$

the error of the reading as an estimate of the position before the mark, the delivered impulse, and the displacement of the position.

Proposition 1. Always $[\hat{N}, \hat{D}] = -i\hbar - [\hat{y}, \hat{D}] - [\hat{N}, \hat{p}]$. If $\hat{N}$, $\hat{D}$ and $\hat{X}$ commute with $\hat{y}$ and $\hat{p}$, then

$$[\hat{N}, \hat{D}] = -i\hbar, \quad [\hat{X}, \hat{N}] = [\hat{X}, \hat{D}] = 0,$$

so $\delta^2\Delta^2 - c^2 \geq \hbar^2/4$ with c the symmetrized covariance of error and impulse, for every joint state.

Proof. $\hat{R}$ acts on the apparatus and $\hat{p}$ on the body, so $[U^\dagger \hat{R} U, U^\dagger \hat{p} U] = U^\dagger [\hat{R}, \hat{p}] U = 0$, that is $[\hat{y} + \hat{N}, \hat{p} + \hat{D}] = 0$; expanding with $[\hat{y}, \hat{p}] = i\hbar$ gives the identity. Under the hypothesis the cross terms vanish. The same argument with $[U^\dagger \hat{R} U, U^\dagger \hat{y} U] = 0$ gives $[\hat{N}, \hat{X}] = 0$, and the preservation of the canonical relation, $[\hat{y} + \hat{X}, \hat{p} + \hat{D}] = i\hbar$, gives $[\hat{X}, \hat{D}] = 0$. The last statement is the Robertson–Schrödinger inequality. $\square$

Commutation with $\hat{y}$ and $\hat{p}$ is meant in the strong sense, with their spectral projections, equivalently with the Weyl operators. Operators commuting in that sense with $\hat{y}$ and $\hat{p}$ commute with every body operator, so the hypothesis says exactly that $\hat{N}$, $\hat{D}$, $\hat{X}$ are apparatus operators.

2. The class is the momentum-transfer class

Proposition 2. Suppose $\hat{X} = 0$ and $\hat{N}, \hat{D}$ are apparatus operators. Then $U = (\mathbf{1} \otimes U_0) e^{-i\hat{y} \otimes \hat{B}/\hbar}$ for an apparatus unitary U_0 and a self-adjoint apparatus operator $\hat{B}$, the delivered impulse is $\hat{D} = -\hat{B}$, and $\hat{R}_0 = U_0^\dagger \hat{R} U_0$ satisfies $[\hat{R}_0, \hat{B}] = i\hbar$, with $\hat{N} = \hat{R}_0$.

Proof. $U^\dagger \hat{y} U = \hat{y}$ means U commutes with $\hat{y}$, so in the position representation $U = \int^\oplus U(y) dy$ with apparatus unitaries $U(y)$, measurable in y . Exponentiate the two hypotheses. Since $\hat{D}$ commutes with $\hat{p}$, $U^\dagger e^{ia\hat{p}/\hbar} U = e^{ia(\hat{p} + \hat{D})/\hbar} = e^{ia\hat{p}/\hbar} e^{ia\hat{D}/\hbar}$, and in the decomposition, where $e^{ia\hat{p}/\hbar}$ translates y by a , this reads $U(y)^\dagger U(y + a) = e^{ia\hat{D}/\hbar}$ for almost every y and every a . The measurable solutions are $U(y) = U_0 e^{iy\hat{D}/\hbar}$, so $U = (\mathbf{1} \otimes U_0) e^{-i\hat{y} \otimes \hat{B}/\hbar}$ with $\hat{B} = -\hat{D}$, and no differentiability is needed. For the pointer, $\hat{y}$ commutes with $\hat{N}$, so $U^\dagger e^{ib\hat{R}} U = e^{ib\hat{y}} e^{ib\hat{N}}$; in the decomposition, with $\hat{R}_0 = U_0^\dagger \hat{R} U_0$, this reads $e^{iy\hat{B}/\hbar} e^{ib\hat{R}_0} e^{-iy\hat{B}/\hbar} = e^{iby} e^{ib\hat{N}}$ for every y and b . At $y = 0$ it gives $\hat{N} = \hat{R}_0$, and for every y it is the Weyl relation between $\hat{B}$ and $\hat{R}_0$, which contains $[\hat{R}_0, \hat{B}] = i\hbar$. $\square$

By the Stone–von Neumann theorem, which applies to the Weyl relation just obtained, the pair $(\hat{R}_0, \hat{B})$ is a canonical pair on one factor of the apparatus, $L^2(\mathbb{R}) \otimes \mathcal{K}$, and $\mathcal{K}$ carries whatever else the apparatus holds. So a mark that leaves the position alone and whose noise is independent of the body is a von Neumann mark: it couples the body’s position to one canonical momentum of the apparatus and reads the conjugate coordinate. Theorem A and the recoil theorem apply to it verbatim, with $\lambda \hat{P}_A$ replaced by $\hat{B}$ and the probe’s state replaced by its reduced state on the canonical factor, entangled with $\mathcal{K}$ or not; Theorems B and C apply with their Gaussian hypothesis placed on that reduced state.

3. Marks that also displace the body

A mark with $\hat{X} \neq 0$ adds the jump $\hat{X}_j$ to the body’s position at t_j , and every later reading carries it. Proposition 1 makes $\hat{X}_j$ commute with the pair $(\hat{N}_j, \hat{D}_j)$, so it acts on the multiplicity factor $\mathcal{K}$. An apparatus state that entangles $\mathcal{K}$ with the canonical factor can correlate $\hat{X}_j$ with the error and the impulse, so the Gaussian floors of the mark-cost note need that correlation bounded too. The recoil bound needs nothing more.

Proposition 3. Theorem R of the recoil note holds for every protocol of marks whose $\hat{N}_j, \hat{D}_j, \hat{X}_j$ are apparatus operators, with Δ_j the spread of $\hat{D}_j$.

Proof. Lemma 2 of the recoil note needs a unitary on apparatus j that translates $\hat{N}_j$ by c_j and leaves everything else in the record unchanged. The record depends on apparatus j through $\hat{N}_j, \hat{D}_j$ and $\hat{X}_j$ only. The unitary $e^{ic_j \hat{D}_j/\hbar}$, applied to the initial state of apparatus j , translates $\hat{N}_j$ by c_j because $[\hat{N}_j, \hat{D}_j] = -i\hbar$ (for the von Neumann mark it is the translation of $\hat{Q}_j$ by $\lambda_j c_j$), and fixes $\hat{D}_j$ and $\hat{X}_j$, which commute with $\hat{D}_j$. Mandelstam–Tamm bounds its effect on the state by $|c_j| \Delta_{\hat{D}_j}/\hbar$, and the rest of the proof of Theorem R is unchanged. An apparatus state entangled across marks is covered too: the $\hat{D}_j$ commute, so the translations combine into one unitary generated by $\sum_j c_j \hat{D}_j$, and the seminorm property $\Delta(\sum_j c_j \hat{D}_j) \leq \sum_j |c_j| \Delta_{\hat{D}_j}$ gives the same bound. $\square$

4. Consequence for STATE

Option 1 is settled. Body-independent noise forces the canonical pair, the position-undisturbed marks with such noise are exactly the von Neumann marks, and the recoil bound $s \sum_j \Delta_j \geq 8(1 - 2\epsilon)\hbar$ covers every mark with body-independent noise, position-displacing ones included. What remains outside is noise that depends on the body, where $[\hat{y}, \hat{D}]$ or $[\hat{N}, \hat{p}]$ survives in Proposition 1; that is Ozawa’s setting and option 2.